Tableau Training 02

1. Data Sources in Tableau

1. **Connect to Data Sources**
    - Tableau supports connections to **file systems** (CSV, Excel), **relational systems** (SQL Server, Oracle), and **cloud systems** (Azure, BigQuery).
    - It also supports **ODBC** connections for other data sources.
2. **Connect Live vs In-Memory**
    - **Connect Live**: Real-time data connection. Results are continuously updated.
    - **In-Memory**: Data is cached in memory for faster analysis.
3. **Combine Data Sources**
    - Supports multiple connections for **data blending**.

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2. Extracting Data

1. **Creating Extracts**
    - Menu Path: **Data → Extract Data**.
    - Apply filters to limit rows or aggregate dimensions.
2. **Applying Extract Filters**
    - Use **Add Filter** to include specific rows or sub-categories.
3. **Append Data**
    - Add new data to an existing extract: **Data → Extract → Append Data from File**.
4. **Extract History**
    - View the history of data extracts: **Data → Extract History**.

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3. Field Operations

1. **Adding Fields**
    - Drag and drop fields into **Columns**, **Rows**, or other shelves in the worksheet.

2. **Combining Fields**
    - Combine two fields into one with concatenated values.
3. **Searching and Reordering Fields**
    - Use the search box to find fields quickly.
    - Drag fields to reorder their position.

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4. Editing Metadata

1. **Checking Metadata**
    - Tableau displays data source metadata (column names and types).
2. **Changing Data Types**
    - Edit incorrect data types manually.
3. **Renaming and Hiding Columns**
    - Rename fields for clarity. Hide unnecessary fields.
4. **Column Aliases**
    - Assign aliases to improve column understanding.

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5. Data Joining

1. **Creating a Join**
    - Go to **Data → Edit Data Source**.
    - Drag tables into the data pane. Tableau auto-generates a join (editable).
2. **Editing Join Type**
    - Modify the join type (Inner, Left Outer, etc.) by clicking on the join diagram.
3. **Editing Join Fields**
    - Change the fields forming the join conditions.

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6. Working with Worksheets

1. **Adding and Renaming Worksheets**
    - Add new worksheets via right-click or the **New Worksheet** icon.

- Rename worksheets for better organization.

2. **Saving, Deleting, and Reordering Worksheets**
    - Save a worksheet.
    - Delete using right-click → **Delete Sheet**.
    - Reorder by dragging worksheets to a new position.
3. **Creating a Paged Workbook**
    - Place a dimension (e.g., Sub-Category) on the **Pages Shelf** to generate paginated views.
    - Use **Page Controls** for navigation and automatic playback.

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7. Operators and Calculations

1. **Operators**
    - General, Arithmetic (+, -, /), Relational (>, <, =), and Logical (AND, OR, NOT).
2. **Operator Precedence**
    - Operations follow a specific order, with parentheses taking precedence.
3. **Numeric and Aggregate Calculations**
    - Use built-in functions for custom calculations:
        - Example: SUM(Sales), AVG(Profit).
4. **Create Calculated Fields**
    - Path: **Analysis → Create Calculated Field**.
    - Write custom formulas (e.g., subtract discount from profit).
    - Use aggregate functions for summarized calculations.

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8. Functions in Tableau

1. **Number Functions**
    - Examples: CEILING, ROUND, POWER.
2. **String Functions**

- Examples: LEN, UPPER, REPLACE.

3. **Date Functions**
    - Examples: DATEADD, DATENAME, NOW.
4. **Logical Functions**
    - Examples: IFNULL, ISDATE.
5. **Aggregate Functions**
    - Examples: AVG, COUNT, MEDIAN.